

## Suggested Mark Scheme

- 1 In flax plants, the flowers are white, lilac or blue. Fig.1 shows the pathway by which the flower cells produce coloured pigments.

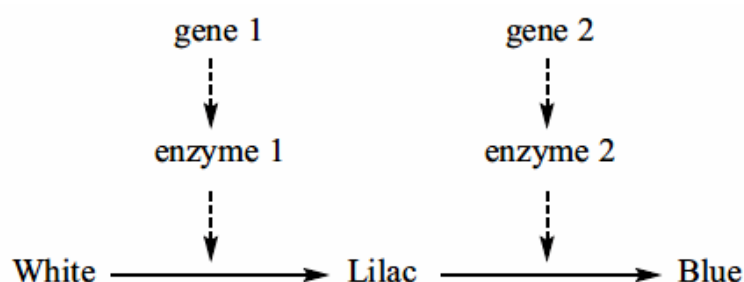


Fig. 1

- (a)(i) A single base deletion occurs in gene 1. Describe and explain the effect of this mutation on the colour of flax plant flowers.

*frameshift mutation;*

*aa seq. of enz 1 changed downstream of deletion;*

*change in R groups;*

*thus change in bonds / R group interactions in polypeptide;*

*leading to different folding thus 3D configuration of enz 1;*

*likely to be non-functional / does not catalyse synthesis of lilac pigment;*

*lack of substrate for enz 2 to act on to synthesize blue pigment;*

*flax flowers appear white;*

[4]

- (b) Gel electrophoresis was used to separate the enzymes involved in this pathway. When extracts of the differently coloured flax petals were analysed, four different patterns of bands were produced. In the table, only bands that contain functional enzymes are shown.

| Result of electrophoresis | Colour of petal |
|---------------------------|-----------------|
|                           | White           |
|                           | blue            |
|                           | lilac           |
|                           | white           |

- (i) With reference to Fig. 1, complete the table to give the colour of the petal from which each extract was taken. [3]
- (ii) Describe how the enzymes were separated by gel electrophoresis.

*polyacrylamide gel electrophoresis;*

*smaller pore size for higher resolution required for prot;*

*denature / linearize prot to remove influence of shape on migration speed;*

*and confer equal negative charge per unit mass using SDS / detergent;*

*load prot sample in wells near the negative electrode / cathode;*

*add buffer for conductivity;*

*apply voltage / electric current across gel;*

*-vely charged prot migrate towards anode;*

*speed / distance of migration is inversely proportional to mass / Mw;*

*large prot more impeded by gel / small prot less impeded;*

*polyacrylamide gel acts as molecular sieve;*

*stain with Coomassie blue / Ag stain to visualize prot bands;* [5]

- (c) Genomic libraries and cDNA libraries can be created for the flax plant genome. State **three** differences between the two gene libraries.

|           | genomic                                                                                        | cDNA                                                                                         |
|-----------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| size      | large fragments containing segments of all genes in entire genome                              | smaller fragments containing only genes expressed in tissue of interest                      |
| source    | shotgun method from cleaving genomic DNA with RE                                               | reverse transcribed from total set of mature mRNA extracted from tissue of interest using RT |
| introns   | presence of introns & non-coding regions (incl regulatory and repetitive seq)                  | absence of introns, only exons;                                                              |
| use       | use to study gene expression due to presence of reg seq like promoters;                        | usually to decipher aa seq / clone to produce polypeptide;                                   |
| vectors   | both need to be ligated into plasmid / phage vectors and transformed into bacterial host;      |                                                                                              |
| screening | both libraries contain more than one gene, thus need to be probed to isolate gene of interest; |                                                                                              |

[3]

[Total: 15]

- 2 Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme ADA.

Some children with SCID have been treated with stem cells. Stem cells can divide and develop into any type of blood cell, including white blood cells. The treatment used with the children is described in Fig. 2.1.

**Fig. 2.1**

- (a)(i) Explain how the lack of a functional gene coding for ADA leads to SCID.  
*non-functional / no ADA produced;*  
*cells unable to deaminate purines;*  
*accumulation of adenosine / deoxyadenosine in lymphocytes cause toxicity;*  
*cause lymphocytes to die / inability to mature thus impair immune*  
*system;* [2]
- (ii) Suggest and explain **two** reasons why stem cells were used in this treatment.  
*treated stems cells have unlimited self-renewal;*  
*and ability to differentiate in lymphocytes;*  
*effects of treatment can thus be sustained;*  
*normal ADA gene can be propagated to daughter cells;* [2]
- (iii) Suggest what stem cells are isolated from the bone marrow.  
*haematopoietic stem cells;* [1]
- (iv) Describe how many copies of functional ADA gene may be produced in the process shown in Fig.2.1.  
*via polymerase chain reaction;*  
*denature at 95°C;*  
*separate dsDNA by breaking H-bonds;*  
*annealing primers at 55°C;*  
*primer complementary base pair to region flanking target sequence;*  
*elongation at 70°C;*  
*Taq polymerase elongates primer at 3' end by*  
*bringing in deoxyribonucleotides complementary to template strand;* [3]
- (v) State **two** advantages of using virus to carry the gene into the stem cells.  
*1 able to deliver genes to specific target tissue;*  
*2 high transfection efficiency;*

*able to integrate genes into genome / delivers gene into nucleus,*

*allows stable propagation / reduce chance of cytoplasmic degradation;* [2]

- (vi) Suggest why the method described in Fig. 2.1 is better than transplanting stem cells with functional ADA gene from a donor.

*stem cell from donor may lead to tissue rejection*

*or invoke immune response;* [1]

- (b) Child A and Child B were treated with genetically engineered stem cells described in Fig. 2.1. Fig. 2.2 shows the number of functioning white blood cells in both children during the year following treatment. Children who do not suffer from SCID have between 5000 and 8000 white blood cells per  $\text{mm}^3$  of blood.

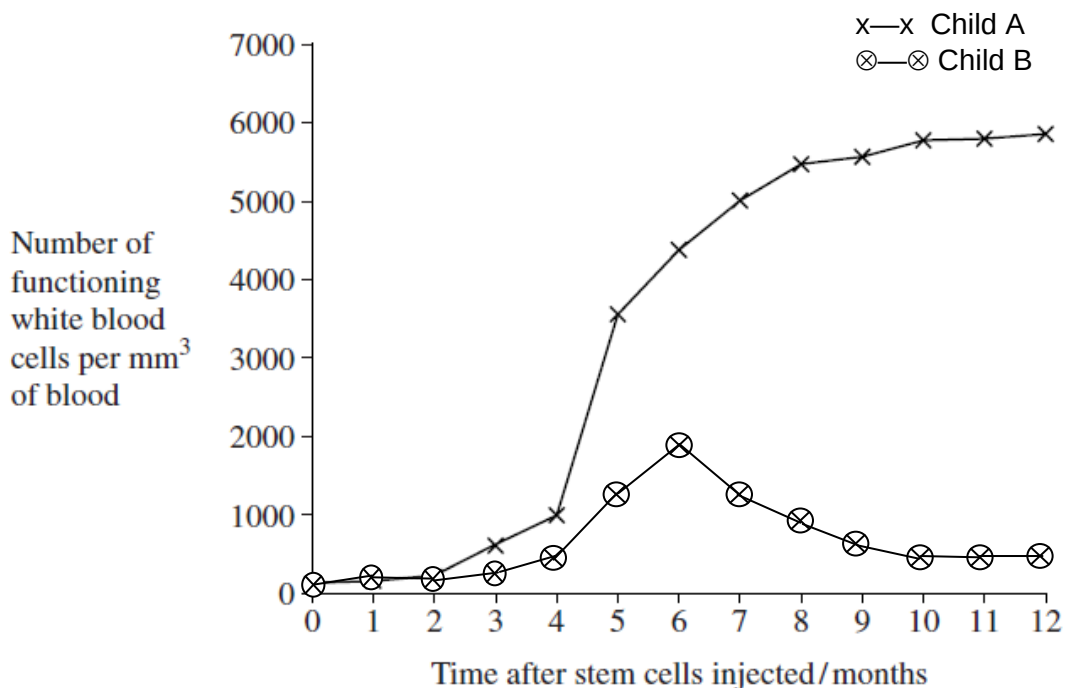


Fig. 2.2

- (i) Describe the results of the treatment.

*A - no. of wbc increased from 100 – 6000 cells/mm³ of bld from 0 – 8 mths;  
stabilize at 6000 cells/mm³ fr 8 – 12 mths aft treatment;*

*B – increase fr 100 – 2000 fr 0 – 6 mths;*

*decrease fr 2000 – 500 fr 6 – 12 mths;* [2]

- (ii) Explain the difference in response between the two children to treatment.

*gene successfully integrated into stem cell genome; stably propagated;*

*gene failed to integrate stem cell genome in Child B;*

*not stably propagated to daughter lymphocytes / degraded;*

*or death of treated stem cell;* [2]

[Total: 15]

- 3 Genetic engineering is a technique that has been developed to improve the quality and yield of many crop plants and animals.

Numbers of genetically modified (GM) plants can be increased by cloning from tissue culture.

- (a) What do you understand by 'genetically modified (GM) plants'?  
*plants with foreign gene from same/diff species/org artificially integrated into their genome;*

*foreign gene is expressed by org to form a new/novel trait;* [1]

- (b) Describe the process of cloning such plants.

*obtain explant & sterilize with bleach;*

*grow explant in nutrient medium of mineral salts, sucrose, aas, vitamins, plt growth regulators (auxin & cytokinin);*

*work under aseptic condns;*

*placed in tissue culture room under controlled condns with optimal temp + light + humidity;*

*vary conc. of plt growth regulators to stimulate cell division & differentiation into callus, shoot or root growth;*

*plantlets formed transplanted to soil in greenhouse;*

*once matured, moved to harsher env gradually;* [3]

Maize can be genetically modified to produce a toxin, which acts as an insecticide, killing any insects which feed on the plants.

Growth of these genetically modified maize plants suggested that the toxin might also kill a range of beneficial soil organisms, such as insects, worms, fungi and bacteria.

To investigate this, seeds of the maize were surface sterilized and germinated on sterile agar. The seedlings were then grown at a constant temperature in both sterile and non-sterile soil-free nutrient medium in closed containers.

After 6, 12 and 18 days the plants were transferred to fresh medium. The concentration of the toxin in the medium was measured. The larvae of several insect species were put in contact with it. The results were summarized in Table 3.1.

**Table 3.1**

| time/days | 6       | 12      | 18      | 6           | 12          | 18          |
|-----------|---------|---------|---------|-------------|-------------|-------------|
| medium    | sterile | sterile | sterile | not sterile | not sterile | not sterile |

|                                                           |    |    |    |   |   |   |
|-----------------------------------------------------------|----|----|----|---|---|---|
| toxin in medium/ a.u.                                     | 8  | 10 | 12 | 1 | 0 | 0 |
| % mortality of insects after 5 days of exposure to medium | 90 | 95 | 99 | 0 | 0 | 0 |

(c) With reference to Table 3.1,

(i) describe a suitable control for this experiment

*germinate surface-sterilised seeds of non-GM maize on sterile agar;  
seedlings grown at a same const temp in a sterile, soil-free  
nutrient medium in closed containers like GM counterparts;* [1]

(ii) describe and explain the results under sterile conditions.

*toxin in medium increases from 6 to 18 days;  
toxin level increases from 8 to 12 a.u.;  
increase is caused by pdtn of toxin by GM maize;  
%mortality of insects increases from 6 to 18 days;  
% mortality increases from 90% to 99%;  
toxin produced must have killed insect larvae feeding on GM  
maize;* [3]

(iii) discuss the extent to which the results confirm that growing the genetically modified maize will damage the beneficial soil organisms.

*unlikely;  
toxin secreted by GM maize into soil is likely to be broken down by  
m/org present in soil;  
in non-sterile medium with only 1 a.u. of toxin left from 6 to 18 days;  
low conc of toxin is non-lethal to insects, 0% larval mortality at 1 a.u. of  
toxin;  
likely to;  
although there are m/org that break down toxin, larvae on or feeding  
on GM maize leaves will still die;  
in sterile medium with 8 – 10 a.u. of toxin, 95% mortality;  
may topple ecological balance if insects are preys to higher  
consumers in soil;* [3]

(d) Discuss briefly the ethical and social implications of genetically modified organism.

*might cause allergies in humans after consumption, as foreign  
antigens;  
may be toxic/ carcinogenic endangering human safety;  
transgenes from GM plts may transfer to wild type plts thru cross-*

*pollination/escaped, killing non-target insects, emergence of superweeds;*

*may lead to reduction in biodiversity/upset ecological balance, toxic to beneficial insects feeding on GM maize;*

*may cause leaching of toxin to rivers & contamination of water source;*

*devts may benefit rich countries at expense of poorer countries;*

*World food pdtn may become dominated by small no. of large countries;*

[4]

**2 ethical & 2 social implications need to be stated with elaboration for each.**

[Total: 15]

#### Question 4

- (a) Describe the functions of **three** enzymes needed in the formation of recombinant DNA. [6]

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① restriction endonuclease

1. to cleave target gene and plasmid / vector by hydrolyzing phosphodiester bonds at specific recognition site;
2. to produce sticky ends to hold plasmid and target gene temporarily via H bonds;
3. facilitate catalysis of phosphodiester bond formation by ligase by holding target gene and vector in correct orientation and close proximity;

② ligase

4. catalysed formation of phosphodiester bonds btw target gene and plasmid;
5. to seal sugar-phosphate backbone;

③ terminal transferase

6. add multiple DNA bases / poly(dN) tails to 3' end of target gene and vector;
7. complementary bases added to target gene and plasmid;
8. create sticky ends for DNA cut by RE producing blunt ends;
9. AVP;

(@ 1m each)

- (b) Describe how nucleic acid hybridization can be used in RFLP analysis on isolated DNA sample. [6]

- 
1. digest isolated DNA sample with specific RE at specific RE producing restriction fragments;
  2. restriction fragments of different length and nos. due to diff location and no. of RE sites in DNA sample;
  3. separate restriction fragments by gel eletrophoresis according to size/molecular wt;
  4. restriction fragments in electrophoresized gel transferred to adhere to nitrocellulose paper;
  5. using sponge and paper towel placed on top of gel to wick buffer through gel;
  6. NaOH in buffer denature dsDNA into ssDNA;
  7. expose to radioactive probe, a short DNA/RNA seq complementary to target gene;
  8. probe hybridize to restriction fragment with complementary base sequence via H-bonding;
  9. excess probe is washed off;



- 
10. expose to photographic film to produce autoradiograph;
  11. target DNA show up as dark bands, position of bands allow identification of length;
  12. when compared to DNA ladder;
  13. banding pattern, in terms of fragment no. and length provides info for RFLP analysis;
  14. known as Southern blotting;
  15. AVP;

(@ ½ m each)

(c) Explain the application of RFLP analysis in disease detection and DNA fingerprinting. [8]

- 
1. RFLP is based on variation in no. & length of restriction fragments btw individuals;
  2. Diff patterns (positions & no.) of restriction fragments detected by use of RE;
  3. sorted by gel electrophoresis / nucleic acid hybridization / Southern blotting

#### *Disease detection*

4. disease causing allele differs from normal allele due to mutation;
5. alters ability of RE to recognize & digest at specific site;
6. mutation within restriction site in an allele may cause restriction site to be eliminated;
7. leading to larger fragment & loss of 2 smaller fragments (diff fragment no.);
8. indicates presence / inheritance of specific allele;
9. RFLP marker is closely linked to or within disease causing allele;
10. serves as indicator for disease causing allele;
11. crossing over btw marker & allele unlikely to occur during meiosis;
12. RFLP marker & allele always inherited together;
13. e.g. Huntington disease, sickle cell anaemia, haemophilia etc;

#### *DNA fingerprinting*

14. DNA profiling process identifying individuals based on diff in repetitive DNA seq.;
15. VNTR / lengths of repeated DNA seq in DNA non-coding region;
16. diff no. /size of VNTRs among individuals is unique / highly polymorphic repeats;

- 17. results in different no. of restriction length fragments;*
- 18. VNTRs can be probed for seq flanking it;*
- 19. DNA fingerprint analysed / compared agst another individual to detect matching banding pattern;*
- 20. e.g. paternity testing / forensic analysis for crime investigation / trace evolutionary lineage;*

[Total: 20]